

AJAY VENKATESH

Brooklyn, NY

+1 516 585 9013 av3855@nyu.edu linkedin.com/in/ajay-venkatesh-32013b17b github.com/ajayv-19

Experience

Amazon

May 2025 – Aug 2025

Software Development Engineer Intern

Seattle, US

- Designed a secure UI Console architecture using Midway SSO, OAuth, and HELIS to enforce least-privilege, role-based access; presented the solution to the team and received design approval.
- Built serverless infrastructure using AWS CDK to deploy API Gateway, Lambda (Java 21), DynamoDB, and S3 with cross-account access; implemented CI/CD pipeline for multi-stage deployments.
- Developed the frontend using React 18, integrating AWS UI components, Midway AuthN, and HELIS AuthZ with role-aware views and filtering; delivered a UI that was 40% faster and significantly more secure than older version.
- Implemented backend APIs using Java Coral, Smithy models, Dagger DI, and CBOR serialization; improved response time by 35% and ensured reliability through comprehensive unit and integration testing.

New York University

Aug 2024 – May 2025

Software Developer (On-Campus)

New York, US

- Contributed to an AI-driven Remote Patient Monitoring System, awarded \$100K by the National Science Foundation, USA, employing React Native, TypeScript, and Cognito authentication for real-time health monitoring.
- Authored parallelized ETL pipelines in Python to convert unstructured NoSQL data into structured SQL using GCP Cloud Functions. Leveraged concurrent processing of data chunks, reducing transformation time by 60%.
- Developed a React website to host QuickSight dashboard handling 50K+ records, utilized Cognito, Amplify, and Lambda functions to establish login, role-based access control, and Stripe integration for payment gateway.

PricewaterhouseCoopers (PwC)

Aug 2021 – Aug 2024

Senior Software Engineer

- Built a distributed backend on Microsoft Azure for a health platform, integrating Microsoft Dynamics for secure data flow and using Azure services for scalability, fault tolerance, and event-driven processing.
- Led code reviews & mentored interns on design, TDD, and optimization, reducing bugs by 25% and accelerating CI/CD in an Agile setup.
- Developed a scalable B2B platform with modular services to replace legacy onboarding, integrating pipelines, email tracking, and BI reporting, improving automation by 70%.
- Designed and deployed a cognitive AI solution using OCR and NLP on Azure, integrating it with business workflows through REST APIs, ensuring scalability, security, and compliance with PwC delivery standards.
- Led project delivery across design, development, testing, and deployment, aligning with PwC's SDLC framework, coordinating with PMs, architects, and business stakeholders.
- Implemented RBAC and session validation for 5+ roles, enforcing least-privilege access and preventing unauthorized actions.
- Optimized data pipeline by parallelizing Excel file imports (100K+ rows), cutting processing time from 5 hours to 1 and boosting efficiency by 80%.

Projects

Flight Delay Prediction

- Developed a scalable PySpark ML pipeline to predict airline flight delays, processing over 7M records stored in HDFS.
- Extracted and aggregated historical flight datasets using SQL queries for feature engineering and anomaly detection.
- Conducted extensive exploratory data analysis (EDA) to uncover delay patterns, key predictors, and systemic bottlenecks.
- Trained and optimized Random Forest and Gradient Boosted Tree models, achieving **85%+ accuracy** for delay classification.
- Enabled data-driven operational planning by translating model outputs into actionable insights for delay mitigation and scheduling improvements.

Skills

Programming Languages: Python, Java, C++, JavaScript (ES6), SQL

Web Development: REST API, GraphQL API, Microservices, React, Node.js, Express.js, Sequelize

Database: RDBMS (MySQL, PostgreSQL), NoSQL (MongoDB, JSON), Dataverse

Cloud: AWS (CDK, SDK, EC2, S3, DynamoDB, API Gateway, Lambda, Cognito, IAM, Q, SageMaker), Azure (AI Services, Azure AI Foundry, Cognitive Services, Logic Apps, Function Apps, Storage Blobs, Security Data Lakes)

AI/ML: Neural Networks, Classification, Pattern Recognition, Data Extraction (OCR/NLP), RAG, MLOps, AI Solution Deployment, Responsible AI, Model Monitoring, Vector Search, AI Governance, Model Validation & Monitoring

Automation Platforms: Microsoft Copilot Studio, Power Automate, Power Apps, Power BI, Azure AI Studio, Google ADK

Tools: Git, Docker, Jupyter, Visual Studio, Postman, QuickSight, CI/CD Pipelines

Methodology: Agile, Scrum, SDLC Governance, Test-Driven Development (TDD)

Education

New York University | Master of Science in Computer Engineering (GPA – 3.9/4)

Aug 2024 – Aug 2026

Certifications

AWS Cloud Practitioner

Microsoft Developer Associate

Power BI Data Analyst Associate

Microsoft Solution Architect Expert

Azure Data Scientist Associate

Azure Data Fundamentals

Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques (*IEEE*)

An Aquila-optimized SVM classifier for Diabetes prediction (*IEEE*)